Lecture 3 v2:
Simple Linear Regression

25117 - Econometrics

Universitat Pompeu Fabra

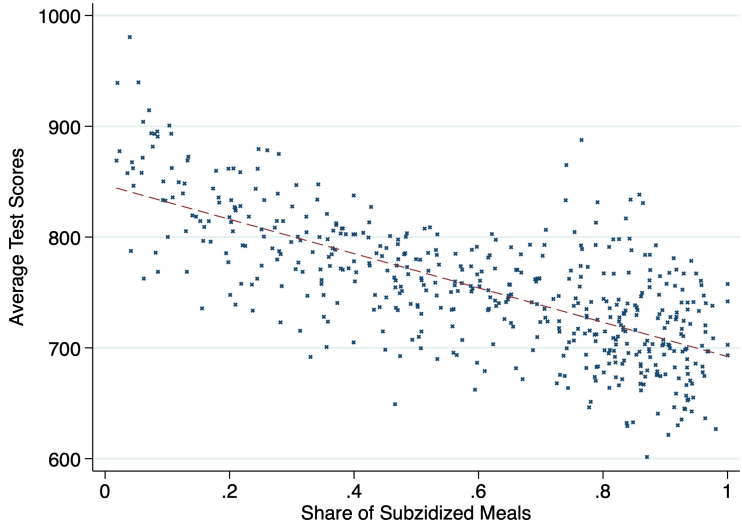
October 7th, 2024

What we learned in the last lesson

- The sample average, $\overline{Y}$, is an estimator of the population mean, μ_Y. When $Y_1, \ldots, Y_n$ are i.i.d.,
  1. The sampling distribution of $\overline{Y}$ has mean μ_Y and variance $\sigma^2_{\overline{Y}} = \sigma^2_Y/n$
  2. $\overline{Y}$ is unbiased
  3. by the LLN, $\overline{Y}$ is consistent; and
  4. by the CLT, $\overline{Y}$ has an approximately normal sampling distribution when the n is large.
- The t-statistic is used to test the null hypothesis that the population mean takes on a particular value. If n is large, the t-stat $\sim \mathcal{N}(0,1)$ under H_0.
- The t-statistic can be used to calculate the p-value associated with the null hypothesis. The p-value is the probability of drawing a statistic at least as adverse to the null hypothesis as the one you actually computed in your sample, assuming the null hypothesis is correct. A small p-value is evidence that the null hypothesis is false.
- A 95% confidence interval for μ_Y contains the true value of μ_Y in 95% of all possible samples.
- The sample correlation coefficient is an estimator of the population correlation coefficient and measures the linear relationship between two variables – that is, how well their scatterplot is approximated by a straight line.

The regression line

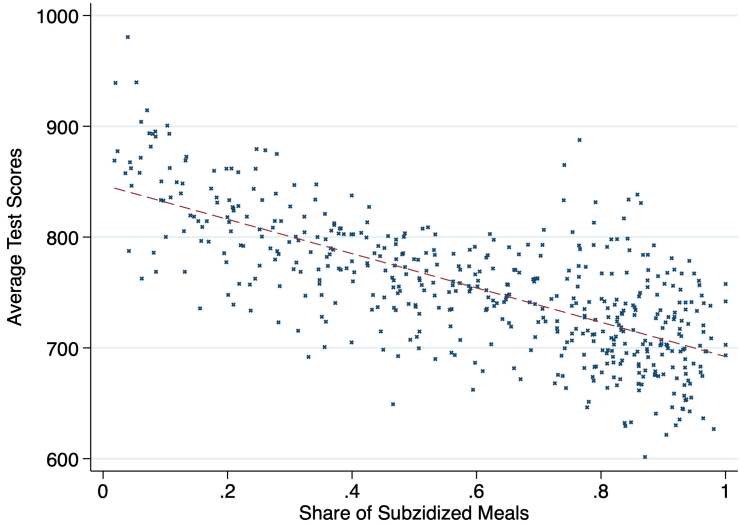
- The population regression line is the expected value of Y given X.
- The slope is the difference in the expected values of Y, for two values of X that differ by one unit



The regression line

The estimated regression can be used either for

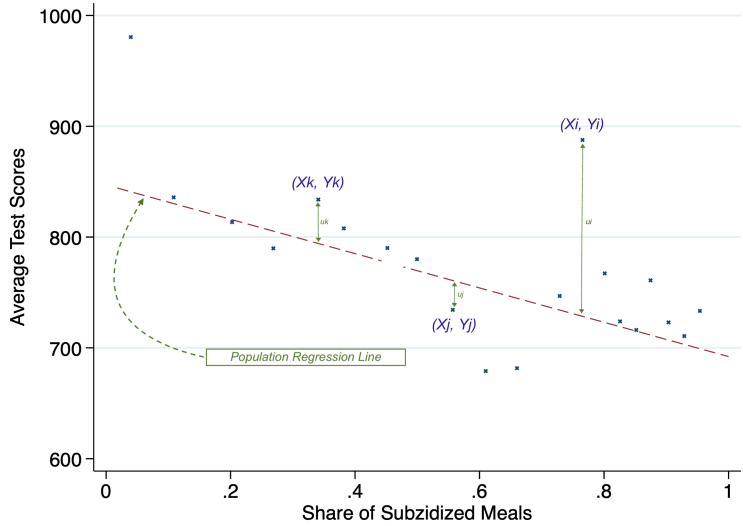
1. **causal inference** (learning about the causal effect on Y of a change in X)
   - → What is the causal impact of subsidizing meals on students' performances?
2. **prediction** (predicting the value of Y given X, for an observation not in the data set)
   - → What are the students' expected test scores given the share of subsidized meals?



The Population Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + U_i$$

- There are n observations, (X_i, Y_i), $i = \{1, \ldots, n\}$
- Y is the dependent variable
- X is the independent variable or regressor
- $\mathbb{E}[Y \mid X] = \beta_0 + \beta_1 X$ is the population regression line
- β_0 is called the intercept
- β_1 is the slope
- U_i is the random error; u_i is its realization (as plotted).



The Ordinary Least Squares (OLS) Estimator

How can we estimate β_0 and β_1 from the data? Recall from Lecture 1 that the Least Square estimator of μ_Y, $\overline{Y}$, solves

$$\min_m \sum_{i=1}^{n} (Y_i - m)^2$$

By analogy, the OLS estimators of the unknown parameters β_0 and β_1, $\hat{\beta}_0$ and $\hat{\beta}_1$ solves,

$$\begin{aligned} & \min_{\beta_0,\beta_1} \sum_{i=1}^{n} U_i^2 \\ \Leftrightarrow \quad & \min_{\beta_0,\beta_1} \sum_{i=1}^{n} (Y_i - \mathbb{E}[Y \mid X])^2 \\ \Leftrightarrow \quad & \min_{\beta_0,\beta_1} \sum_{i=1}^{n} (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2 \end{aligned}$$

The Ordinary Least Squares (OLS) Estimator

- The first-order conditions are

$$\sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \qquad \sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)X_i = 0.$$

- **Step 1: solve the first condition for the intercept.**

$$n\overline{Y} - n\hat{\beta}_0 - n\hat{\beta}_1\overline{X} = 0 \quad \Longrightarrow \quad \boxed{\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1\overline{X}}.$$

- **Step 2: substitute and simplify the second condition.**

$$\begin{aligned} 0 &= \sum_{i=1}^{n}\left[Y_i - \left(\overline{Y} - \hat{\beta}_1\overline{X}\right) - \hat{\beta}_1 X_i\right] X_i \\ &= \sum_{i=1}^{n}\left[(Y_i - \overline{Y}) + \hat{\beta}_1\overline{X} - \hat{\beta}_1 X_i\right] X_i \\ &= \sum_{i=1}^{n}\left[(Y_i - \overline{Y}) - \hat{\beta}_1(X_i - \overline{X})\right] X_i. \end{aligned}$$

The Ordinary Least Squares (OLS) Estimator

$$\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}, \qquad \hat{\beta}_1 = \frac{\sum_{i=1}^{n}(Y_i - \overline{Y})(X_i - \overline{X})}{\sum_{i=1}^{n}(X_i - \overline{X})^2} = \frac{s_{XY}}{s_X^2}.$$

- The intercept adjusts the average outcome for the fitted slope.
- The slope compares the comovement of X and Y with the variation in X.

Back to the Linear Regression Model

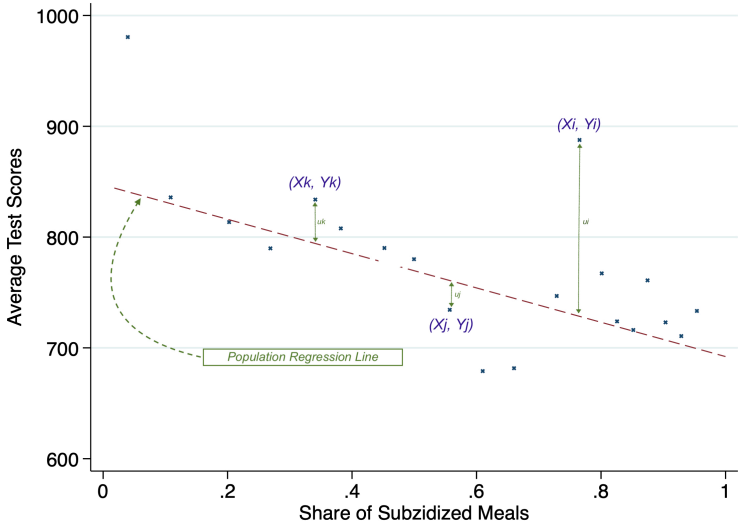
The OLS estimator minimizes the average squared difference between the actual values of i Y and the prediction ("predicted value") based on the estimated line.

- $\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}$
- $\hat{\beta}_1 = \frac{s_{XY}}{s_X^2}$
- $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$
- $\hat{U}_i = Y_i - \hat{Y}_i$

This means that

- $\Delta \mathbb{E}[Y_i \mid X_i] = \hat{\beta}_1 \Delta X_i$

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Interpretation

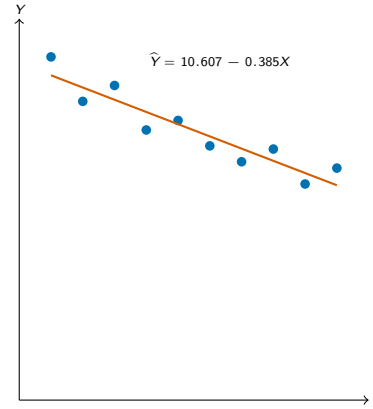
Here, the population regression line is given by

$$\hat{\beta}_0 \approx 754 + 155 \times .6 \approx 847$$
$$\hat{\beta}_1 \approx \frac{-11.78}{.28^2} \approx -155$$
$$\hat{Y}_i \approx 847 - 155 \times X_i$$

- This means that a 10ppt increase in the share of subsidized meals is associated with ($\neq$ cause!) a 15.5 decrease in the average test scores.
- To give a standard of comparison, this is about 26% of the Test scores' standard deviation!
- The intercept (taken literally) means that, according to this estimated line, districts with zero meals would have a (predicted) test score of approximately 847...
- But this interpretation of the intercept makes no sense – it extrapolates the line outside the range of the data (see Lect2 for descriptive stats) – here, the intercept is not economically meaningful.

Example: Estimating an OLS Line with 10 Observations

i	X_i	Y_i	$X_i - \overline{X}$	$(X_i - \overline{X})^2$	$Y_i - \overline{Y}$	$(X_i - \overline{X})(Y_i - \overline{Y})$
1	1	10.8	-4.5	20.25	2.31	-10.40
2	2	9.4	-3.5	12.25	0.91	-3.19
3	3	9.9	-2.5	6.25	1.41	-3.53
4	4	8.5	-1.5	2.25	0.01	-0.01
5	5	8.8	-0.5	0.25	0.31	-0.16
6	6	8.0	0.5	0.25	-0.49	-0.25
7	7	7.5	1.5	2.25	-0.99	-1.49
8	8	7.9	2.5	6.25	-0.59	-1.47
9	9	6.8	3.5	12.25	-1.69	-5.92
10	10	7.3	4.5	20.25	-1.19	-5.36
Sums:	55	84.90	0	82.50	0	-31.75



Calculation

$$\overline{X} = 5.5, \quad \overline{Y} = 8.49,$$

$$s_{XY} = \frac{-31.75}{9} = -3.528, \quad s_X^2 = \frac{82.50}{9} = 9.167,$$

$$\hat{\beta}_1 = \frac{s_{XY}}{s_X^2} = -0.385, \quad \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X} = 10.607.$$

Goodness of Fit

- How well does our regression model fit the data? This is a crucial question.
- Goodness of fit measures help us assess the quality of our model.
- Two regression statistics provide complementary measures of how well the regression line "fits" or explains the data:
  1. The regression R^2 measures the fraction of the variance of Y that is explained by X; it is unitless and ranges between zero (no fit) and one (perfect fit)
  2. The standard error of the regression (SER) measures the magnitude of a typical regression residual in the units of Y.
- These measures are very useful to predict Y when X is known but Y is not (out-of-sample prediction). A common prediction measure is $\hat{Y} \pm SER$.

R^2

From the previous slides, we note that

$$Y_i = \hat{Y}_i + \hat{U}_i$$

The R^2 is the ratio of the sample variance of $\hat{Y}$ to the sample variance of Y

$$\begin{aligned} R^2 &= \frac{\text{explained sum of squares } (ESS)}{\text{total sum of squares } (TSS)} = \frac{\sum_{i=1}^{n}(\hat{Y}_i - \overline{Y})^2}{\sum_{i=1}^{n}(Y_i - \overline{Y})^2} \\ &= 1 - \frac{\text{sum of squared residuals } (SSR)}{\text{total sum of squares } (TSS)} = 1 - \frac{\sum_{i=1}^{n}\hat{U}_i^2}{\sum_{i=1}^{n}(Y_i - \overline{Y})^2} \end{aligned}$$

R^2 ranges from 0 to 1, where higher values indicate a better fit (i.e., the regressor is good at predicting the outcome)

The Standard Error of the Regression (*SER*)

1. Population errors:

$$U_i = Y_i - (\beta_0 + \beta_1 X_i).$$

If $\mathbb{E}[U_i] = 0$, then

$$\sigma_U^2 = \mathrm{Var}(U_i) = \mathbb{E}[U_i^2], \text{ and } \sigma_U = \sqrt{\mathbb{E}[U_i^2]}.$$

σ_U measures the typical spread around the population regression line, in units of Y.

2. Observable residuals: The true coefficients are unknown, so replace them by their estimates:

$$\hat{U}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i).$$

The intercept first-order condition gives

$$-2\sum_{i=1}^{n} \hat{U}_i = 0, \text{ hence } \overline{\hat{U}} = 0.$$

3. Estimate the error variance and standard deviation:

$$\hat{\sigma}_U^2 = \frac{1}{n-2}\sum_{i=1}^{n}(\hat{U}_i - \overline{\hat{U}})^2 = \frac{SSR}{n-2}; \text{ and } SER = \hat{\sigma}_U = \sqrt{\hat{\sigma}_U^2}.$$

The $n-2$ denominator corrects for estimating the intercept and slope.

Under the usual homoskedastic OLS assumptions, SER^2 is unbiased for σ_U^2; SER estimates σ_U.

As with the sample standard deviation s_Y, we estimate an unknown spread from deviations around a fitted mean; see the comparison next.

From the Sample Standard Deviation to SER

Before: unknown σ_Y

We want to estimate the population variance:

$$\sigma_Y^2 = \mathbb{E}[(Y_i - \mu_Y)^2].$$

Estimate the mean μ_Y by $\overline{Y}$: **one fitted parameter**, hence $n-1$.

$$s_Y^2 = \frac{\sum_{i=1}^n (Y_i - \overline{Y})^2}{n-1}, \qquad s_Y = \sqrt{s_Y^2}.$$

s_Y estimates the outcome spread σ_Y.

Regression: unknown σ_U

We want to estimate the error variance:

$$\sigma_U^2 = \mathbb{E}[U_i^2], \text{ since } \mathbb{E}[U_i] = 0.$$

Estimate the mean $\beta_0 + \beta_1 X_i$ by $\hat{Y}_i$: **two fitted parameters**, hence $n-2$.

$$\hat{U}_i = Y_i - \hat{Y}_i, \text{ and } SER^2 = \frac{\sum_{i=1}^n \hat{U}_i^2}{n-2}.$$

$SER = \sqrt{SER^2}$ estimates the error spread σ_U.

The comparison: s_Y estimates the spread of outcomes around μ_Y; SER estimates the spread of regression errors around the regression line.

SER and RMSE: What Is the Difference?

Both measures summarize the typical size of the OLS residuals.

- First define the residual sum of squares:

$$SSR = \sum_{i=1}^{n} \hat{U}_i^2.$$

- The *SER* divides by $n - 2$:

$$SER = \sqrt{\frac{SSR}{n-2}}.$$

  The two degrees of freedom are used to estimate β_0 and β_1.

- The fitted value is $\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i$, so the prediction error is

$$Y_i - \widehat{Y}_i = \widehat{U}_i.$$

  Therefore, the *RMSE* is the root mean squared prediction error:

$$RMSE = \sqrt{\frac{SSR}{n}}.$$

  It is the average size of the errors $Y_i - \widehat{Y}_i$ in the observed sample.

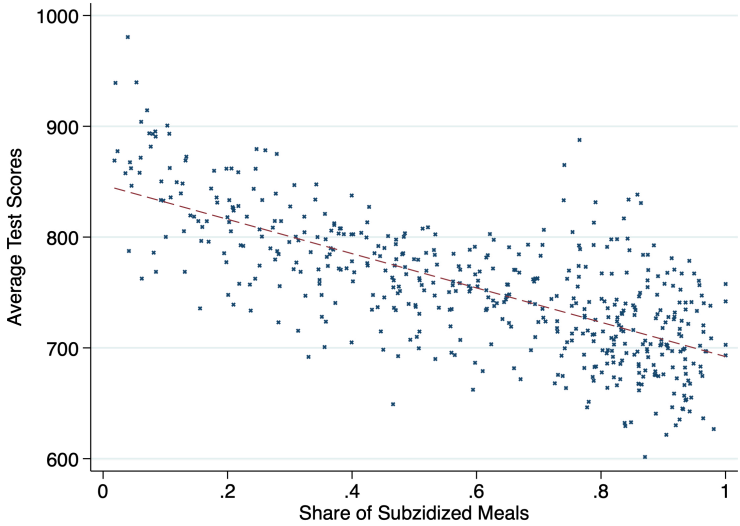
- For a large n, $n - 2$ and n are almost the same, so *SER* and *RMSE* are very similar.

Interpretation

Both are measured in the units of Y. They describe the typical size of the part of Y that the regression does not explain.

Goodness of Fit

- Here the $R^2 = .509$ and $RMSE = 42.601$
- About half of the variation in actual test scores is explained by the variation in the predicted test scores (which, in our regression specification, comes from the variation in subsidized meals).
- Does this make sense? Does this mean that subsidizing school meals matters for education in a policy sense?



Population OLS: FOCs and Results

In the population, choose $(\hat{\beta}_0, \hat{\beta}_1)$ to minimize the mean squared prediction error:

$$Q(\hat{\beta}_0, \hat{\beta}_1) = \mathbb{E}\left[(Y - \hat{\beta}_0 - \hat{\beta}_1 X)^2\right].$$

The population first-order conditions are

$$0 = \frac{\partial Q}{\partial \hat{\beta}_0} = -2\mathbb{E}[Y - \hat{\beta}_0 - \hat{\beta}_1 X],$$
$$0 = \frac{\partial Q}{\partial \hat{\beta}_1} = -2\mathbb{E}[X(Y - \hat{\beta}_0 - \hat{\beta}_1 X)].$$

The first condition gives

$$\hat{\beta}_0 = \mathbb{E}[Y] - \hat{\beta}_1 \mathbb{E}[X].$$

Substituting this into the second condition gives the population OLS coefficients:

$$\hat{\beta}_1 = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}, \qquad \hat{\beta}_0 = \mathbb{E}[Y] - \hat{\beta}_1 \mathbb{E}[X].$$

The conditions for causality

The OLS Assumptions for Causal Inference

Let

$$Y_i = \beta_0 + \beta_1 X_i + U_i \;\; ; \;\; i = 1, ..., n$$

where β_1 is the causal effect of X on Y. Then:

1. The error term U_i has conditional mean 0 given X_i: $\mathbb{E}[U_i \mid X_i] = 0$;
2. (X_i, Y_i) $i = 1, ..., n$, are independent and identically distributed (i.i.d.) draws from their joint distribution; and
3. Large outliers are unlikely: X_i and Y_i have nonzero finite fourth moments.

The conditions for causality

Why do these conditions help us estimate a causal effect?

1. **Causal interpretation:** $\mathbb{E}[U_i \mid X_i] = 0$ means that, once X_i is fixed, the remaining determinants of Y_i in U_i are not systematically related to X_i. Therefore, the variation in Y associated with X identifies β_1 rather than a confounding factor.
2. **Reliable estimation:** i.i.d. sampling makes the observations representative of the population and prevents dependence across observations from distorting the OLS calculation.
3. **Reliable inference:** finite moments and no large outliers ensure that no single observation dominates the estimate and allow the usual large-sample results for standard errors, tests, and confidence intervals.

Do you think these assumptions hold in our California Schools Experiment?

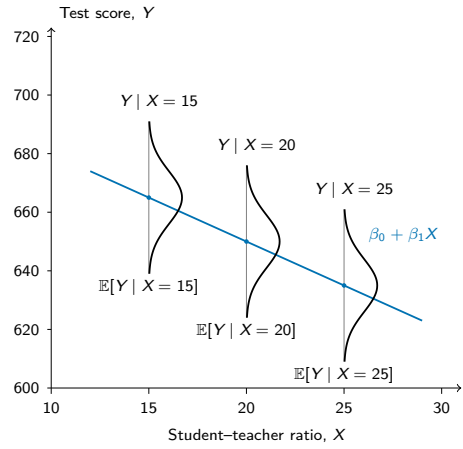
Do the assumptions hold in the California Schools Experiment?

1. **Conditional mean zero: likely questionable.** Let Z_i measure students' economic disadvantage and suppose
$$Y_i = \beta_0 + \beta_1 X_i + \gamma Z_i + V_i, \text{ and } \mathbb{E}[Z_i \mid X_i = 1] \neq \mathbb{E}[Z_i \mid X_i = 0].$$
The regression we estimate omits Z_i and writes
$$Y_i = \beta_0 + \beta_1 X_i + U_i.$$
Comparing this equation with the true equation above,
$$\beta_0 + \beta_1 X_i + U_i = \beta_0 + \beta_1 X_i + \gamma Z_i + V_i,$$
so, after subtracting $\beta_0 + \beta_1 X_i$ from both sides,
$$U_i = \gamma Z_i + V_i,$$
$$\mathbb{E}[U_i \mid X_i = x] = \gamma \mathbb{E}[Z_i \mid X_i = x] + \mathbb{E}[V_i \mid X_i = x].$$
If $\mathbb{E}[V_i \mid X_i = x] = 0$, then
$$\mathbb{E}[U_i \mid X_i = 1] - \mathbb{E}[U_i \mid X_i = 0] = \gamma \left[\mathbb{E}[Z_i \mid X_i = 1] - \mathbb{E}[Z_i \mid X_i = 0]\right] \neq 0.$$
Thus, the zero conditional-mean assumption fails.
2. **i.i.d. sampling: plausible with care.** If schools are sampled randomly from the same population, the observations can be treated as i.i.d. draws. If schools are clustered by district or selected through a non-random process, this assumption needs more justification.
3. **No influential outliers: must be checked.** A school with an unusually large test score or subsidized-meal rate could strongly affect the fitted line and the estimated effect.

The main threat to a causal interpretation is the first assumption: differences between schools may be correlated with both meal subsidies and test scores.

Assumption (1) : $\mathbb{E}[U_i \mid X_i] = 0$

- If $\mathbb{E}[U_i \mid X_i] \neq 0$, it implies that there is systematic variation in Y not explained by X.
- This unexplained variation can confound our causal interpretations, making it challenging to isolate the true causal effect of X on Y.



Assumption (1) : $\mathbb{E}[U_i \mid X_i] = 0$

Then

$$\begin{aligned}\hat{\beta}_1 &= \frac{cov(X, [\beta_0 + \beta_1 X + U])}{var(X)} \\ &= \frac{cov(X, \beta_0)}{var(X)} + \frac{cov(X, \beta_1 X)}{var(X)} + \frac{cov(X, U)}{var(X)} \\ &= \beta_1 + \frac{cov(X, U)}{var(X)}\end{aligned}$$

If $\mathbb{E}[U_i \mid X_i] \neq 0$, then $cov(X, U) \neq 0$... In the California Schools Experiment, schools subsidizing meals (vs. not) are likely to welcome students from poor (vs. rich) economic and social backgrounds – biasing estimates downwards!

Assumption (2): i.i.d. draws

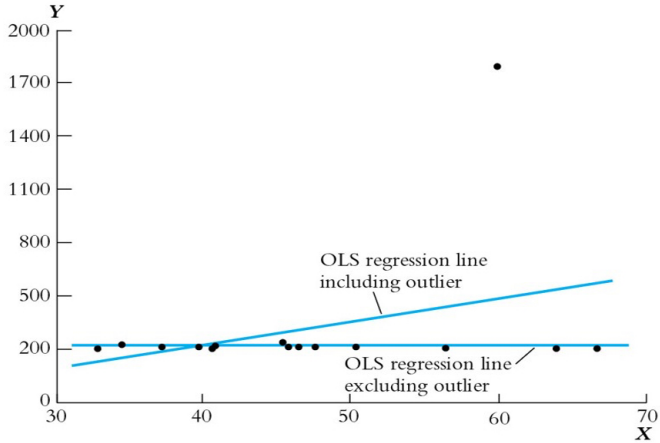
This arises automatically if the entity (individual, district) is sampled by simple random sampling:

- The entities are selected from the same population, so (X_i, Y_i) are identically distributed for all $i = 1, ..., n$.
- The entities are selected at random, so the values of (X, Y) for different entities are independently distributed.

The main place we will encounter non-i.i.d. sampling is when data are recorded over time for the same entity (panel data and time series data) – we will deal with that complication when we cover panel data.

Assumption (3): No large outliers

- If X and Y are bounded, then they have finite fourth moments. (Standardized test scores automatically satisfy this; family income, etc. satisfy this too.)
- The substance of this assumption is that a large outlier can strongly influence the results
- Why is it an outlier? Is it a typo? In practice, outliers are often data glitches (coding or recording problems). If so, they really shouldn't be in your data set!



A conditional-expectation rule

When conditioning on X, any function of X is already known:

$$\mathbb{E}[XY \mid X] = X\mathbb{E}[Y \mid X].$$

Fix $X = x$. Then x is a constant inside the conditional expectation:

$$\mathbb{E}[XY \mid X = x] = \mathbb{E}[xY \mid X = x] = x\mathbb{E}[Y \mid X = x].$$

Because this holds for every possible value x, we can write the result using the random variable X.

OLS Finite Sample Properties

Start from the OLS slope formula and substitute $Y_i = \beta_0 + \beta_1 X_i + U_i$, using $\sum_i (X_i - \overline{X}) = 0$:

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2} = \frac{\sum_i (X_i - \overline{X}) Y_i}{\sum_i (X_i - \overline{X})^2} = \frac{\sum_i (X_i - \overline{X})(\beta_0 + \beta_1 X_i + U_i)}{\sum_i (X_i - \overline{X})^2}.$$

Because $\sum_i (X_i - \overline{X}) = 0$, the intercept term disappears. For the β_1 term, use $X_i = \overline{X} + (X_i - \overline{X})$:

$$\sum_i (X_i - \overline{X}) X_i = \sum_i (X_i - \overline{X}) \left[\overline{X} + (X_i - \overline{X}) \right] = \overline{X} \sum_i (X_i - \overline{X}) + \sum_i (X_i - \overline{X})^2 = \sum_i (X_i - \overline{X})^2.$$

Therefore,

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (X_i - \overline{X}) U_i}{\sum_{i=1}^n (X_i - \overline{X})^2}.$$

1. Conditional unbiasedness: hold X fixed and use $\mathbb{E}[U_i \mid X] = 0$.

$$\mathbb{E}[\hat{\beta}_1 \mid X] = \beta_1 + \frac{\sum_i (X_i - \overline{X}) \mathbb{E}[U_i \mid X]}{\sum_i (X_i - \overline{X})^2} = \beta_1 + 0 = \boxed{\beta_1}.$$

2. Unconditional unbiasedness: apply the law of iterated expectations.

$$\mathbb{E}[\hat{\beta}_1] = \mathbb{E}\left[\mathbb{E}[\hat{\beta}_1 \mid X] \right] = \mathbb{E}[\beta_1] = \boxed{\beta_1}.$$

OLS Finite Sample Properties

Reminder: SER estimates σ_U, so SER^2 estimates σ_U^2.

Write $\hat{\beta}_1 - \beta_1$ as a weighted sum of errors. Conditional on X, the weights are fixed; independence makes the variance of the sum equal to the sum of variances. Under homoskedasticity, $var(U_i \mid X) = \sigma_U^2$.

Define $TSS_X = \sum_{i=1}^n (X_i - \overline{X})^2$, the **total sum of squares of X**: total variation of the observed X_i values around their sample mean.

$$var(\hat{\beta}_1 \mid X) = var(\hat{\beta}_1 - \beta_1 \mid X) = var\left[\frac{\sum_{i=1}^n (X_i - \overline{X})U_i}{TSS_X}\middle| X\right]$$

$$= \frac{1}{TSS_X^2} var\left[\sum_{i=1}^n (X_i - \overline{X})U_i \middle| X\right]$$

$$= \frac{\sum_{i=1}^n (X_i - \overline{X})^2 var(U_i \mid X)}{TSS_X^2} = \frac{\sigma_U^2 \sum_{i=1}^n (X_i - \overline{X})^2}{TSS_X^2} = \frac{\sigma_U^2}{TSS_X}.$$

Plug in the estimated error variance:

$$\widehat{\text{Var}}(\hat{\beta}_1 \mid X) = \frac{SER^2}{TSS_X} = \frac{SSR}{(n-2)TSS_X}; \qquad \widehat{SE}(\hat{\beta}_1) = \frac{SER}{\sqrt{TSS_X}}.$$

Connection to the sample mean: $TSS_X = (n-1)s_X^2$, so it grows approximately in proportion to n when the spread of X stays similar.

$$\widehat{SE}(\hat{\beta}_1) = \frac{SER}{s_X\sqrt{n-1}}, \text{ compared with } \widehat{SE}(\overline{Y}) = \frac{s_Y}{\sqrt{n}}.$$

With similar regressor spread and error variance, the slope's standard error also decreases roughly as $1/\sqrt{n}$: quadrupling n approximately halves it.

OLS Finite Sample Properties

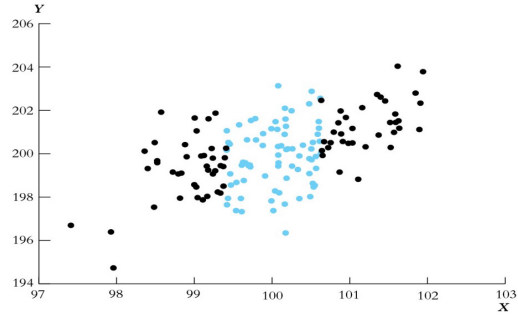
- **Hold the observed X values fixed.** Under homoskedasticity,

$$\text{Var}(\hat{\beta}_1 \mid X) = \frac{\sigma_U^2}{TSS_X}.$$

- Estimate the unknown σ_U^2 by $SER^2 = SSR/(n-2)$:

$$\widehat{\text{Var}}(\hat{\beta}_1 \mid X) = \frac{SER^2}{TSS_X} = \frac{SSR}{(n-2)TSS_X}.$$

- Larger error variance increases the slope's conditional variance: there is more noise around the regression line.



- Holding error variance fixed, more variation in the observed X values increases TSS_X and reduces $\text{Var}(\hat{\beta}_1 \mid X)$.
- With similar regressor spread and error variance, a larger sample increases TSS_X and reduces the slope's conditional variance.
- Ideally, we would choose X to be as spread out as possible – this is sometimes possible with experimental data, but in practice, we often work with random samples

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Let $\mathbf{X} = (X_1, \ldots, X_n)$ denote the full regressor sample, $\bar{U} = n^{-1} \sum_i U_i$, and $TSS_X = \sum_i (X_i - \bar{X})^2 > 0$. Assume $\mathbb{E}[U_i \mid \mathbf{X}] = 0$ for every i.

1. **Average the population equation** $Y_i = \beta_0 + \beta_1 X_i + U_i$:
$$\bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{U}.$$
2. **Substitute into the OLS intercept**:
$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = \beta_0 + \bar{U} - \bar{X}(\hat{\beta}_1 - \beta_1).$$
3. **Condition on X**, so $\bar{X}$ is fixed:
$$\begin{aligned} \mathbb{E}[\hat{\beta}_0 \mid \mathbf{X}] &= \beta_0 + \mathbb{E}[\bar{U} \mid \mathbf{X}] - \bar{X}(\mathbb{E}[\hat{\beta}_1 \mid \mathbf{X}] - \beta_1) \\ &= \beta_0 + 0 - \bar{X} \cdot 0 = \beta_0. \end{aligned}$$
Here $\mathbb{E}[\bar{U} \mid \mathbf{X}] = n^{-1} \sum_i \mathbb{E}[U_i \mid \mathbf{X}] = 0$; conditional unbiasedness of the slope was derived earlier.

By the law of iterated expectations, $\mathbb{E}[\hat{\beta}_0] = \mathbb{E}[\mathbb{E}[\hat{\beta}_0 \mid \mathbf{X}]] = \beta_0$.

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Assume conditionally uncorrelated errors and homoskedasticity: $\mathrm{Var}(U_i \mid \mathbf{X}) = \sigma_U^2$. Write $d_i = X_i - \bar{X}$. Substituting the slope decomposition gives

$$\hat{\beta}_0 - \beta_0 = \bar{U} - \bar{X}\frac{\sum_i d_i U_i}{TSS_X} = \sum_i a_i U_i, \text{ where } a_i = \frac{1}{n} - \frac{\bar{X} d_i}{TSS_X}.$$

Conditional on $\mathbf{X}$, the weights a_i are fixed and cross covariances vanish:

$$\mathrm{Var}(\hat{\beta}_0 \mid \mathbf{X}) = \sigma_U^2 \sum_i a_i^2.$$

Expand the squared weights:

$$\sum_i a_i^2 = \frac{n}{n^2} - \frac{2\bar{X}}{n TSS_X} \underbrace{\sum_i d_i}_{=0} + \frac{\bar{X}^2}{TSS_X^2} \underbrace{\sum_i d_i^2}_{=TSS_X}$$

$$= \frac{1}{n} + \frac{\bar{X}^2}{TSS_X}.$$

Therefore,

$$\boxed{\mathrm{Var}(\hat{\beta}_0 \mid \mathbf{X}) = \sigma_U^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{TSS_X} \right) = \frac{\sigma_U^2 \sum_i X_i^2}{n TSS_X}},$$

because $\sum_i X_i^2 = TSS_X + n\bar{X}^2$.

Testing Whether the Intercept or Slope Is Zero

1. Estimate the regression $Y_i = \beta_0 + \beta_1 X_i + U_i$ by OLS. Use $SER = \sqrt{SSR/(n-2)}$ and $TSS_X = \sum_i (X_i - \bar{X})^2$.

2. Test the intercept

$$H_0 : \beta_0 = 0, \text{ against } H_1 : \beta_0 \neq 0.$$

$$SE(\hat{\beta}_0) = SER\sqrt{\frac{1}{n} + \frac{\bar{X}^2}{TSS_X}}.$$

$$t_0^{\text{obs}} = \frac{\hat{\beta}_0}{SE(\hat{\beta}_0)}.$$

Is the population regression line's value at $X = 0$ different from zero?

2. Test the slope

$$H_0 : \beta_1 = 0, \text{ against } H_1 : \beta_1 \neq 0.$$

$$SE(\hat{\beta}_1) = \frac{SER}{\sqrt{TSS_X}}.$$

$$t_1^{\text{obs}} = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}.$$

Is the population regression slope different from zero?

3. For each test separately, use the normal approximation. Under its null, conditional on X, $T_j \approx \mathcal{N}(0,1)$ for large n, using the OLS assumptions and homoskedasticity, for $j = 0, 1$.

$$p_j \approx 2[1 - \Phi(|t_j^{\text{obs}}|)].$$

4. Decision at 5%: reject that null if $|t_j^{\text{obs}}| > 1.96$ (equivalently, $p_j < 0.05$); otherwise, do not reject it.

These are two separate tests at 5% each, not a joint test that both coefficients are zero.